

Lecture 20

STAT 109: Introductory Biostatistics

Lecture 20: Sampling Methods in Practice

Learning outcomes (review and extend)

- Identify a **sampling frame** for a given population.
 - **Carry out simple random sampling** (e.g. using a list and random numbers).
 - **Classify** a sampling method as convenience, simple random, cluster, systematic, stratified, or multi-stage.
 - Discuss the potential for **bias** and how to reduce it; identify ways to increase **precision**.
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Sampling frame

The **sampling frame** is the list or procedure from which we actually select units. For a given population (e.g. “all undergraduates,” “all patients at a clinic”), we need to ask: *What list or mechanism will we use to choose who gets into the sample?* The frame should match the population as closely as possible. Gaps (undercoverage) or duplicates can cause bias.

Example: To sample undergraduates, the frame might be a registrar’s list of all enrolled students (with IDs). “Students in Stat 109” is a poor frame for “all undergraduates.”

Simple random sampling (SRS)

In **simple random sampling**, every possible subset of n units from the population (or from the frame) has the **same chance** of being the sample. Equivalently, every unit has the same probability of being selected, and selections are made at random.

How to carry out SRS (by hand or with software):

1. **Get a list** of all units in the sampling frame (numbered 1 to N).
2. **Generate random numbers** in the range 1 to N (e.g. using a random number generator, table, or software).
Draw units corresponding to those numbers. If sampling without replacement, skip repeats until you have n distinct units.
3. **Collect data** from the units in the sampling frame that correspond to the n distinct units from step 2.

Example: To select 5 students from a class of 25, number students 1–25. Use a [random number generator](#) (or similar) set to “integers from 1 to 25,” generate numbers, and collect data from the students with those IDs. If a number repeats, draw again until you have 5 distinct IDs.

Classifying sampling methods

Given a description of how a sample was obtained, **classify** it as one of:

Method	What happens
Convenience	Units chosen because they are easy to access (e.g. “people in this room,” “first 50 visitors”). Often not representative.

Method	What happens
Simple random (SRS)	Every set of n units from the frame has the same chance of being selected; usually via random numbers.
Stratified	Population divided into strata (e.g. by major, by year); a random sample is taken from <i>each</i> stratum. Improves representation of each group.
Cluster	Population divided into clusters (e.g. schools, wards); we randomly select <i>clusters</i> and then observe all (or a random sample of) units in those clusters. Useful when listing individuals is hard but listing clusters is easy.
Systematic	From an ordered list, pick a random start, then take every k th unit (e.g. every 10th name).
Multi-stage	Several stages of random selection (e.g. randomly choose schools, then randomly choose students within those schools).

Examples: Classify, generalizability, and improving the design

For each scenario below: (1) **Classify** the sampling method as **random sampling** (e.g. SRS, stratified, cluster) or **convenience** (or other as appropriate). (2) **Impact on generalizability:** Would results generalize well to the population of interest? Why or why not? (3) **How to reduce bias** and **how to increase precision** (one concrete suggestion for each, if applicable).

Example 1: Campus opinion poll

A student group wants to estimate the proportion of all undergraduates who support building a new recreation center. They stand outside the student union from 11 a.m. to 1 p.m. on two weekdays and ask every 10th person who walks by to complete a short survey. They get 120 responses.

- **Classify:** Is this random sampling or convenience (or something else)? What is the sampling frame?
- **Impact on generalizability:** Who might be over- or under-represented? Can we generalize the sample proportion to *all* undergraduates?
- **Reduce bias:** How could the study be redesigned to reduce bias (e.g. who is included)?
- **Increase precision:** What could be done to increase precision of the estimate?

Example 2: Online course feedback

A university wants to estimate the mean satisfaction score (1–5) for all students who took at least one online course last year. They send an email survey to a **random sample** of 500 students drawn from the registrar’s list of all students who were enrolled in at least one online course; 180 students respond.

- **Classify:** Is the *selection* of the 500 random or convenience? What about the 180 who responded?
- **Impact on generalizability:** Even though the initial 500 were a random sample, might the 180 respondents differ from the 320 non-respondents? How does that affect generalizability?
- **Reduce bias:** What type of bias is most relevant here (e.g. nonresponse)? How could bias be reduced?
- **Increase precision:** How could precision be increased (e.g. sample size, design)?

Example 3: Hospital patient satisfaction

A hospital wants to estimate the proportion of patients who are “very satisfied” with their care. They hand out a satisfaction survey to every patient who is discharged from the maternity ward during a 3-month period and receives

the survey before leaving; about 70% return the completed survey.

- **Classify:** How were patients selected? Is this random sampling or convenience?
 - **Impact on generalizability:** To what population would the sample proportion generalize? To all hospital patients? To all maternity patients?
 - **Reduce bias:** What biases might be present (e.g. undercoverage, nonresponse)? Suggest one way to reduce bias.
 - **Increase precision:** Suggest one way to increase precision.
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Reducing bias and increasing precision (summary)

- **Reduce bias:** Use a frame that covers the population; avoid convenience and voluntary response when possible; use random sampling (SRS, stratified, etc.); minimize nonresponse and design surveys to reduce response bias.
 - **Increase precision:** Increase sample size; use designs that reduce variability (e.g. stratified sampling when groups differ a lot).
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Summary

- A **confidence interval** is only as good as the **sample**. Bad sampling (e.g. convenience for a campus-wide question) can make the interval misleading even if the math is correct.
 - **Sampling frame** = list or procedure used to select units. Should align with the population of interest.
 - **SRS** = every subset of size n equally likely; carried out using a list and random numbers.
 - **Classify** methods as convenience, SRS, stratified, cluster, systematic, or multi-stage.
 - **Bias** (sampling, undercoverage, nonresponse, response, voluntary response) can invalidate inference; **precision** is improved by larger n and sometimes by design (e.g. stratification).
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